Date ___/___/______

1) You are building ---- was not required?

Ans Choice : ESP32

Reasons:

- built in Wi-Fi and Bluetooth
- can send data directly to a mobile app / cloud
- faster processor (240 MHz vs Arduino UNO's 16 MHz)
- More memory and GPIO pins
- suitable for IOT applications

WORKING

1) soil moisture sensor measures moisture.
2) ESP32 reads sensor data
3) If moisture is below threshold, ESP32 turns ON pump through relay.
4) Data sent to mobile application every 5 seconds.

If internet connectivity

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is not required

Arduino UNO can be used bec-ause:

→ simpler

→ lower cost

→ enough for local monitoring and pump control.

2. A smart factory contains 500 sensors --- choose and why?

Ans Comparison

PROTOCOL	RANGE	POWER
Wi-Fi	50-100 m	High
Bluetooth	10-100 m	Low
LoRa	2-15 km	Very low
MQTT	Application protocol	Depends on netw-ork.

Date __/__/____

Protocol	Speed
Wifi	Very High
Bluetooth	Meduim
LoRa	Low
MQTT	Efficient

Best Combination

LoRa + MQTT

why?

→ LoRa connects 500 sensors over 2km.
→ Gateway collects sensor data.
→ MQTT sends data efficiently to cloud/ server.
→ low power consumption and large coverage.

Date __/__/__

3. A temp. sensor ---- ADC or software

Ans Step by step troubleshooting

1) check sensor

- replace with known working sensor
- compare readings

2) check wiring

- loose wires
- broken connection
- wrong pin connections

3) check power supply

- measure voltage using multimeter
- ensure stable power

4) Check ADC

- read raw ADC values
- compare with expected values

Date ___/___/___

5) check software

- verify calibration formula
- print raw data using serial monitor
- remove scaling errors.

CONCLUSION →
by testing one component at a time, the faculty section can be identified.

4) An ESP32-based weather -- battery life?

Ans Hardware modifications

→ use low-power sensors

→ use lithium battery

→ add solar charging if possible

→ use efficient voltage regulator.

Date ___/___/_____

Software modifications →

- deep sleep mode
- wake every 10 - 15 minutes
- turn Wi-Fi ON only during transmission
- reduce OLED display usage
- compress transmitted data.

Result :
power consumption decreases significantly, extending battery life.

5. an automatic braking - - - and choose and why ?

Ans Best choice

CAN bus (controller area network)

Reasons : →
- very low latency
- high reliability

Date __/__/____

- error detection mechanisms
- widely used in automobiles

ALTERNATIVE →

automative ethernet for advanced systems.

6. You need to measure ---- circle length 'a' and height 'b'

Ans PLACEMENT

place sensor :

→ at center of room.
→ mid-height (approximately b/2)
→ away from windows, doors, fans, AC vents, and direct sunlight.

WHY ?

→ represents average air-temperature.

Date ___/___/___

→ minimizes local hot / cold effects.

POOR PLACEMENT EFFECTS →

- near window → higher temperature
- near AC → lower temperature
- Near ceiling → hotter air
- near door → fluctuating readings

7. under what situations ----- a digital one?

Ans situations

1) CONTINOUS MEASUREMENTS
analog output changes smoothly.
2) HIGHER RESOLUTION REQUIREMENTS
not limited by digital quantization
3) FAST SIGNAL CHANGES
no conversion delay.
4) LOWER COST APPLICATIONS
many analog sensors are cheaper.

EXAMPLE

Thermistor measuring temp-erature continously.

8 Draw the complete --- one soluti-on for each.

Ans

Sensor

↓

signal conditioning

↓

ADC

↓

ESP 32

↓

Wi-Fi

↓

router

↓

cloud server

↓

data base

↓

mobile app

↓

user.

Date ___ / ___ / ___

FAILURE POINTS AND SOLUTIONS

failure	Solution
sensor failure	redundant sensor
Wiring fault	shielded cables
ADC error	Calibration
ESP32 Crash	Watchdog Timer
Wi-fi failure	Reconnect logic
Server down	Backup server
Database error	Data replication
App failure	Error handling

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Q) If an attacker ---- do not cite far-fetched ideas.

Ans POSSIBLE ATTACKS

1) Unauthorized access
2) Password attacks
3) Data interception
4) Malware injection
5) Deniel of service (DOS)

Security Improvements:

① strong password
Unique credentials

② WPA3 encryption
Secure Wi-Fi communication

③ HTTPS | TLS
encrypted data transmission

④ Secure firmware updates
signed firmware only.

Date ___ / ___ / ___

(5) firewall and access control
restrict unauthorized devices

(6) disable unused ports
reduce attack surface.

(16) Your ESP32 must --- you solve them?

Ans POSSIBLE CONFLICTS

hardware
- GPIO pin conflicts
- power shortage
- I^2C address conflicts

software
- Timing conflicts
- blocking code
- memory issues

SOLUTIONS

- proper pin planning
- separate power supply.

Date ___/___/___

- non-blocking programming
- freeRTOS tasks
- I²C address management.

11) An ESP32 uploads - - - - - perm-
anently lost

Ans Strategy

① store data in
- EEPROM
- SD card
- Flash Memory

② add timestamp

③ retry connection periodically

④ upload stored data after inter-
net returns.

⑤ confirm successful upload before
deleting local copy.

Date __ / __ / ____

BENEFIT

no important data is lost.

12 your IOT device --- long term reliability?

Ans hardware Improvements

→ industrial- Grade components
→ surge protection
→ proper heat dissipation
→ water proof enclosure
→ quality power supply.

Software improvements
→ watchdog timer
→ error recovery
→ memory leak prevention
→ OTA updates
→ data integrity checks

result:
Improved long term reliability.

Date ___ /___ /______

15 Can an IOT system from "intelligent"?

Ans. yes, an IOT system can be considered smart even without artificial intelligence AI

- smart IOT system can sense its environment, process information and perform actions automatically based on predefined rules.

- for eg → a smart irrigation system can automatically switch on a water pump when soil moisture falls below a certain threshold.

- Similarly, a motion sensor can automatically turn on lights when a person enters a room.

- These systems make decisions based on fixed programming rather than learning from data.

DIFFERENCE
Date ___/___/______

→ smart system
- operates using pre-defined rules and conditions
- performs automatic actions without human intervention
- does not learn from past experi ences
- behaviour remains fixed unless reprogrammed.

→ intelligent system: →
- uses AI and machine learning techniques
- learns from historical data and improves performance over time
- can identify patterns and make predictions
- adapts its behaviour according to changing conditions.

EXAMPLE →

→ a smart Thermostat turns on cooling when temp. exceeds a set value.

Date ___/___/___

→ an intelligent thermostat learns user preferences and predicts when cooling will be needed.

CONCLUSION →
all intelligent systems are smart, but not all smart systems are intelligent.

AI enhances IOT systems by making them adaptive, predictive and capable of learning from experience

14. Compare polling ---- IOT device?

Ans POLLING
- The micro-controller continuously checks the status of sensors and devices
- The CPU remains busy checking whether an event has occurred.
- It is simple to implement but consumes more processing power.

ADVANTAGES →

Date __/__/__

→ easy to understand and program.
→ suitable for simple applications.

DISADVANTAGES

- wastes CPU time
- higher power consumption
- slower response to urgent events

INTERRUPT-BASED PROGRAMMING

→ devices send an interrupt signal when an event occurs
→ The CPU responds immediately and executes the required task.
→ More efficient than polling

ADVANTAGES.

fast response time
Better CPU utilization
Lower power consumption

Date ___ /___ /______

applications of interrupts

- fire alarm systems
- motion detection
- Security systems
- push-button control
- vehicle Safety systems

CONCLUSION

→ polling is suitable for simple systems
→ interrupts are preferred in IOT applications requiring quick response and energy efficiency.

15. Energy engineering - - were justified

Ans Every engineering design involves trade-off b/w cost performance power consumption and reliability.

Date __/__/____

① cost vs accuracy

→ low-cost sensors were used instead of industrial-grade sensors

→ This reduced project cost but slightly affected accuracy

→ The trade off was acceptable because of accuracy was sufficient for the application

② Battery life vs update frequency

→ data was transmitted at intervals instead of continuously

→ This increased battery life and reduced power consumption

→ a small delay in updates were considered acceptable

③ simplicity vs advanced features

Date ___/___/___

→ a simple control algorithm was used instead of AI based autom-ation

→ This reduced complexity and made maintainence easier

→ The system still met all project requirements

CONCLUSION →

- These trade offs helped balance performance, cost and effici-ency

- The decisions were justified becau-se they met the project's objectives while keeping the system practi-cal & economical.

Q16. create a flowchart ______ I-O used in system.

Ans start

↓

read sensor

↓

Date ___/___/___

Wi-fi available ?

Yes

↓ upload data → ↓ success → ↓ delete local copy → ↓ repeat

↓ store data locally → ↓ retry later

component used

- ESP32
- sensor
- flash/ SD storage
- Wi Fi module
- MQTT/ HTTP protocol
- cloud server

Q7 Draw a flowchart how an ESP32 - - - . update fails ?

Date ___/___/___

Ans

start

↓

connect wi Fi

↓

check server for update

↓

update available ?

No

↓ ↓

Run current firmware

↓

download firmware

↓

verify signature

↓

valid ?

No

↓ ↓

Reject update

↓

install firmware

↓

restart device

↓

success ?

Date ___ / ___ / ___

Yes

Run new firmware

↓

rollback to previous firmware

Error handling

→ verify checksum
→ verify digital signature
→ rollback mechanism
→ retry download if interrupted

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